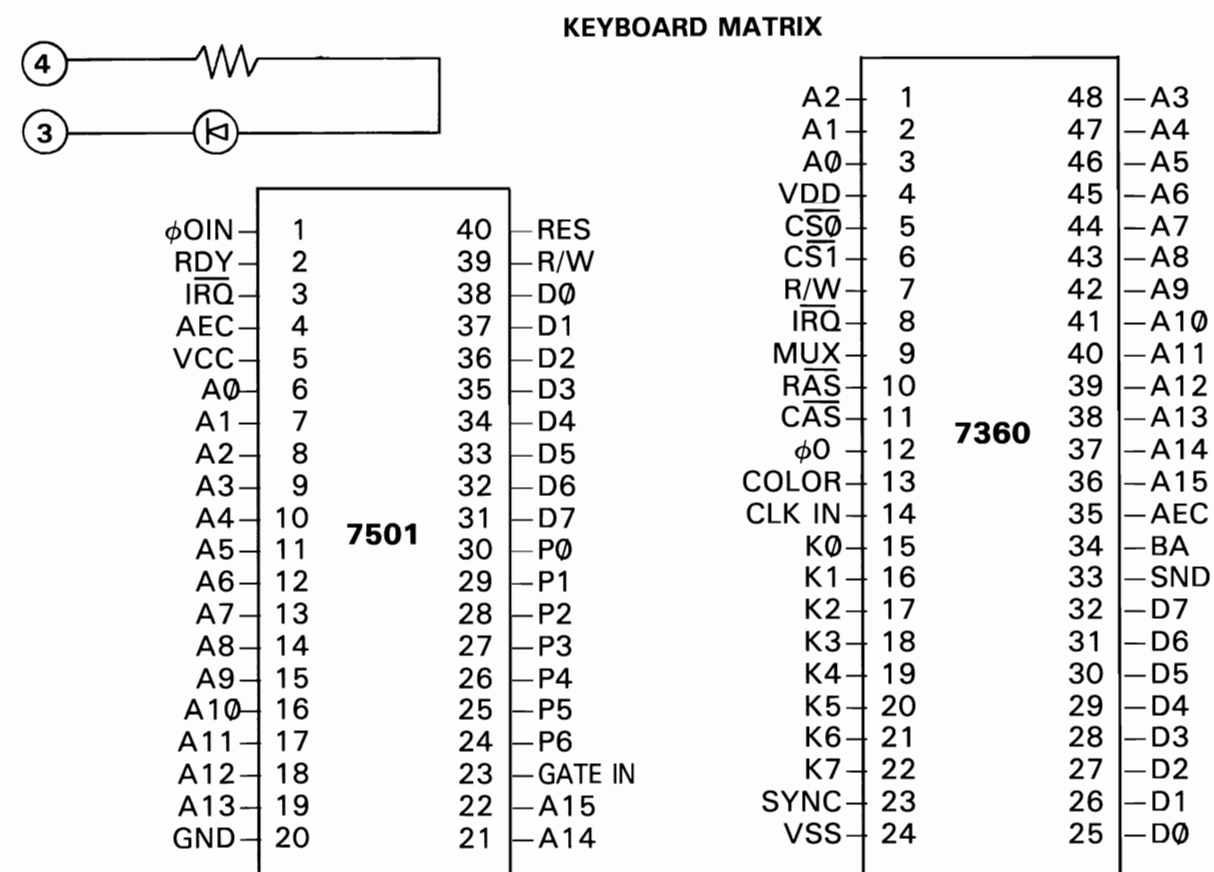
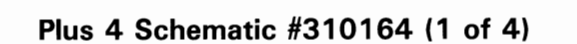
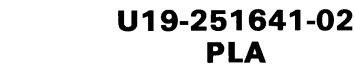
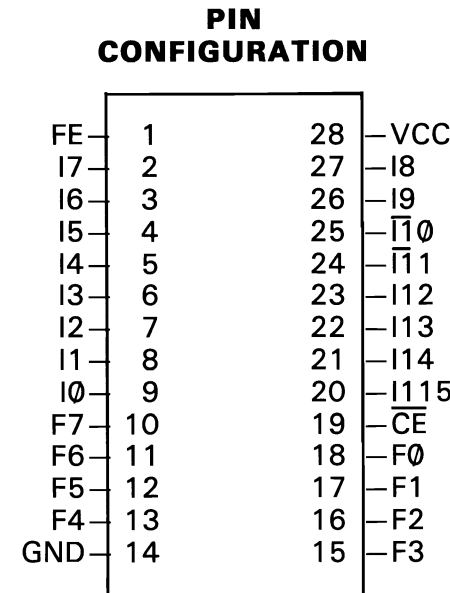
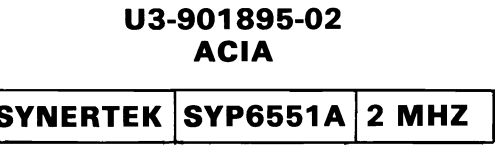
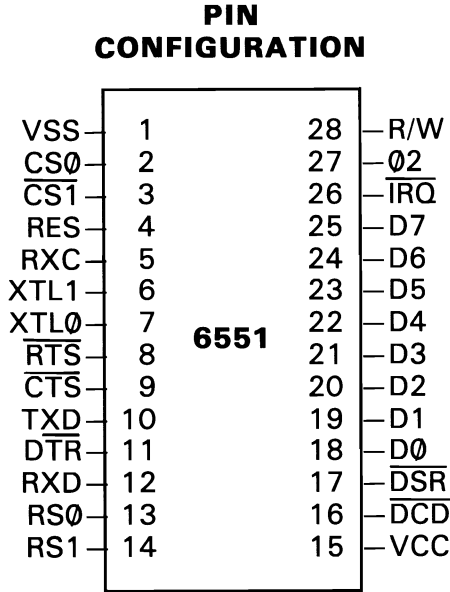




**PIN ASSIGNMENT
U1-251535-01
VLSI, TEXT DISPLAY
(TED)**



M1 SCHEMATIC ON PAGE 11



TRANSMIT/RECIEVE CHARACTERISTICS

CHARACTERISTICS	SYM	-02 2 MHZ		UNIT
		MIN	MAX	
TRANSMIT/RECEIVE CLOCK RATE	t _{CCY}	400	—	ns
TRANSMIT/RECEIVE CLOCK HIGH TIME	t _{CH}	175	—	ns
TRANSMIT/RECEIVE CLOCK LOW TIME	t _{CL}	175	—	ns
XTL1 TO TXD PROPAGATION DELAY	t _{DD}	—	500	ns
RTS PROPAGATION DELAY	t _{DLY}	—	500	ns
IRQ PROPAGATION DELAY (CLEAR)	t _{IRQ}	—	500	ns

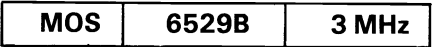
(tr, tf = 10 to 30 ns)

*The Baud Rate with External Clocking is:

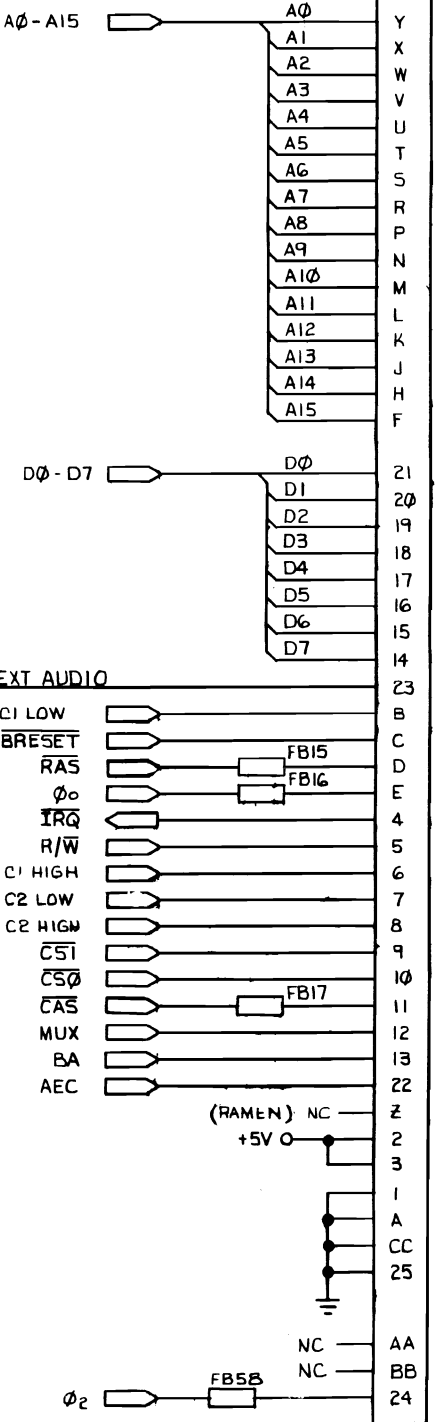
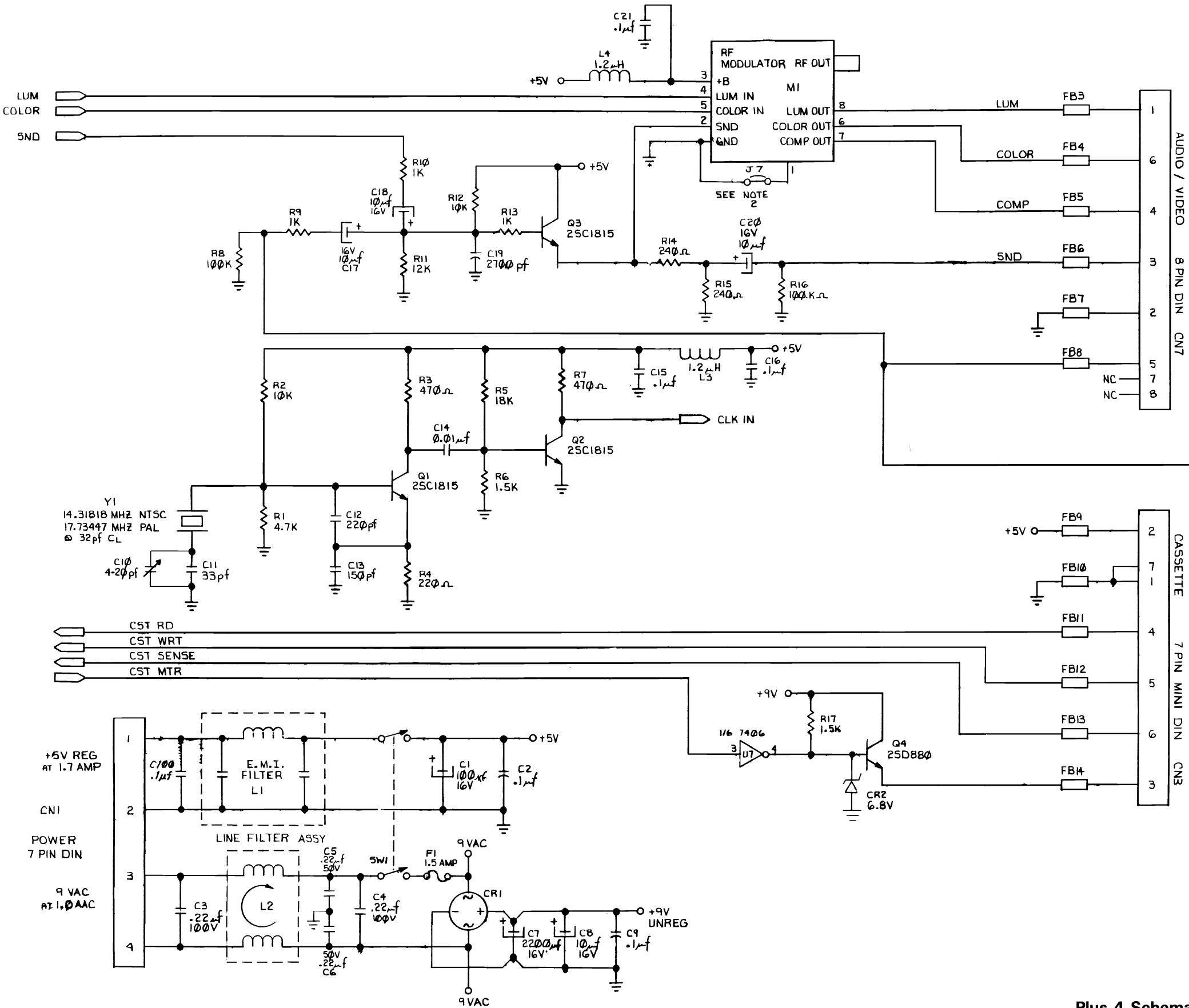
BAUD RATE $\frac{1}{16 \times T_{CCY}}$

CS	R/W	D0-D7
L	L	DATA BUS TO PORT
L	H	PORT TO DATA BUS
H	X	ISOLATION

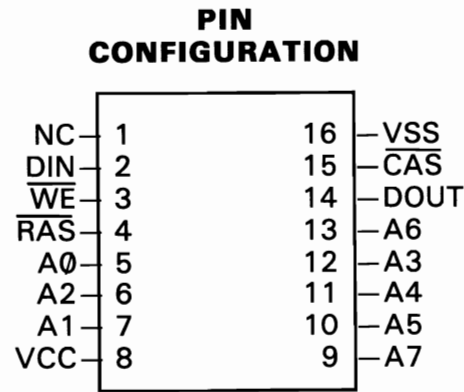
L = LOW LEVEL
H = HIGH LEVEL
X = IRRELEVANT



U5/U27-251640-03
SINGLE PORT
INTERFACE

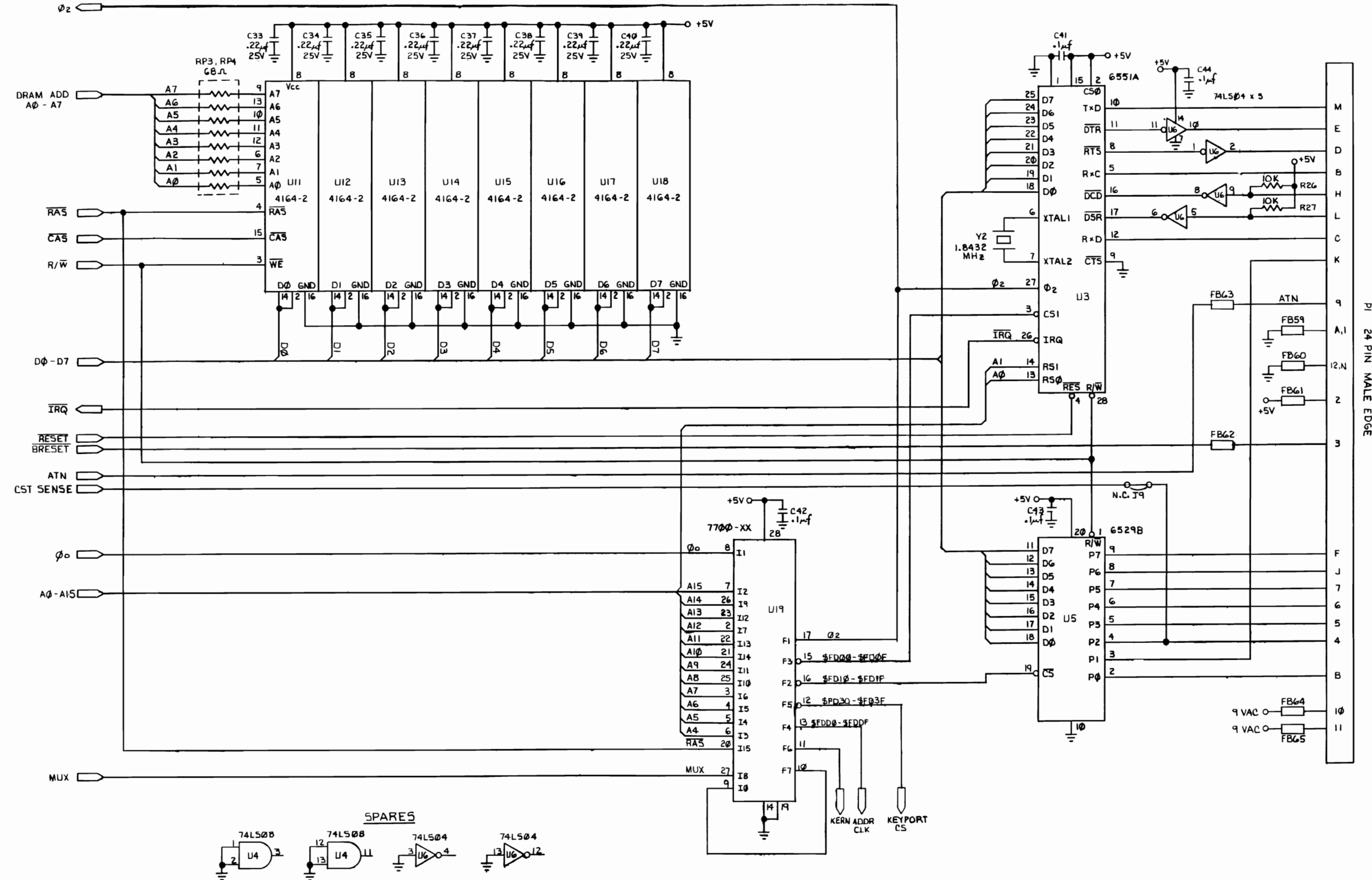
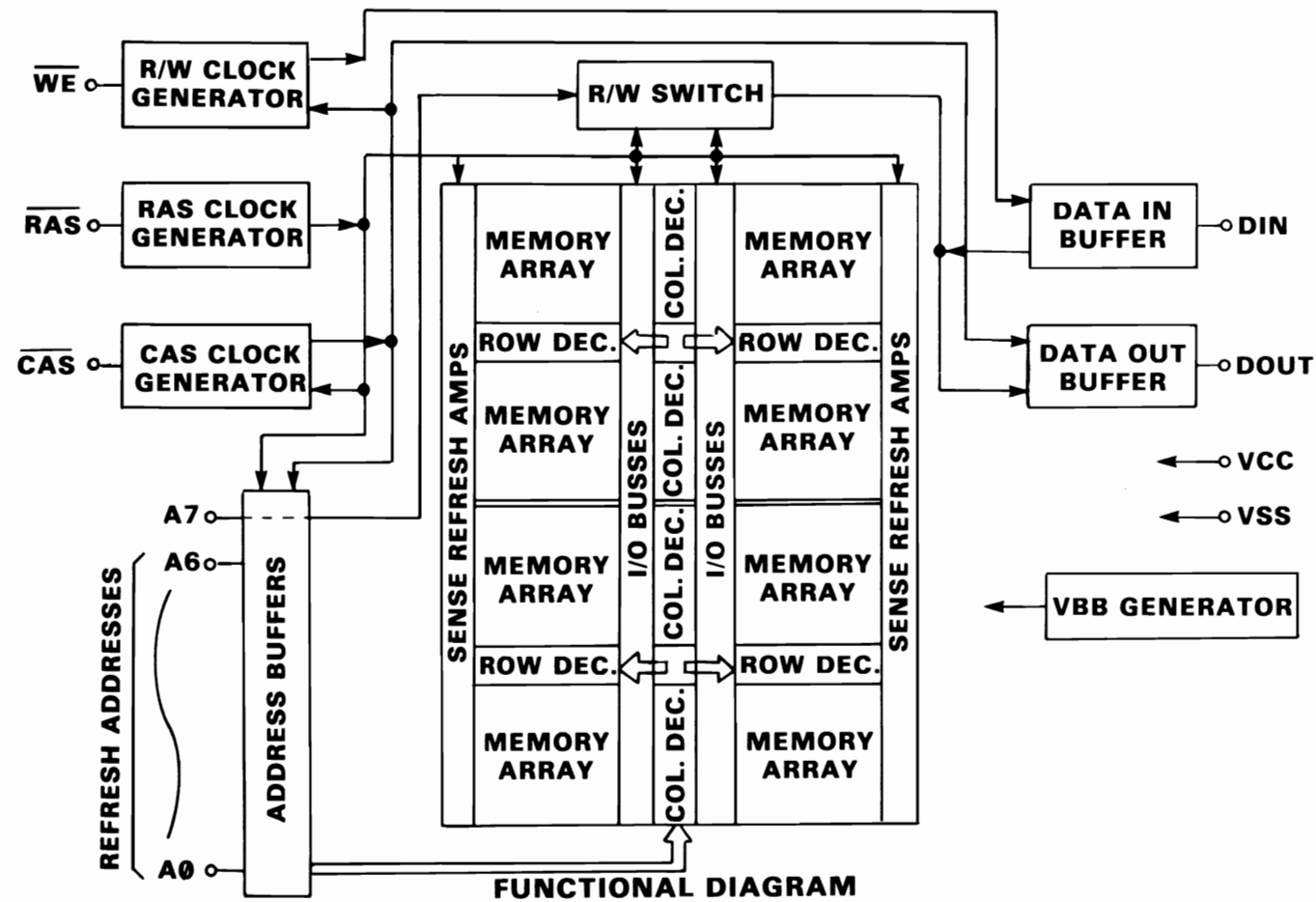


U3, U5, U19 PINOUTS ON PAGE 8



U11 ~ 18
64K DYNAMIC RAM
901505-01

COMMODORE PART NUMBER	APPROVED SOURCE 1 OF SUPPLY	VENDOR PART NUMBER	ACCESS TIME (ns)	CYCLES (ns)	POWER	
					ACTIVE (MW)	STANDBY (MAX)(MW)
901505-01	HITACHI	HM4864-3	200	335	330	20
901505-01	NEC	μPD4164-2	200	375	250	28
901505-01	MITSUBISHI	M5K416NS-20	200	330	275	28
901505-01	MOSTEK	MK4564N-20	200	345	300	22
901505-01	OKI	MSM3764-20	200	330	248	23
901505-01	MICRON TECHNOLOGY	MT4264-3	200	385	300	30
901505-01	HITACHI	HM4864P-3	200	335	330	20
901505-01	MATSUSHITA (PANASONIC)	MN4164P-20	200	330	275	27.5
901505-01	SIEMENS	HYB4164-3	200	330	150	20
901505-01	SHARP	LH2164-Z1	200	330	248	28
901505-01	HITACHI	HM4864AP-3	200	330	242	20
901505-01	TOSHIBA	TMM4164AP-20	200	330	275	22



Plus 4 Schematic #310164 (3 of 4)

PIN CONFIGURATION

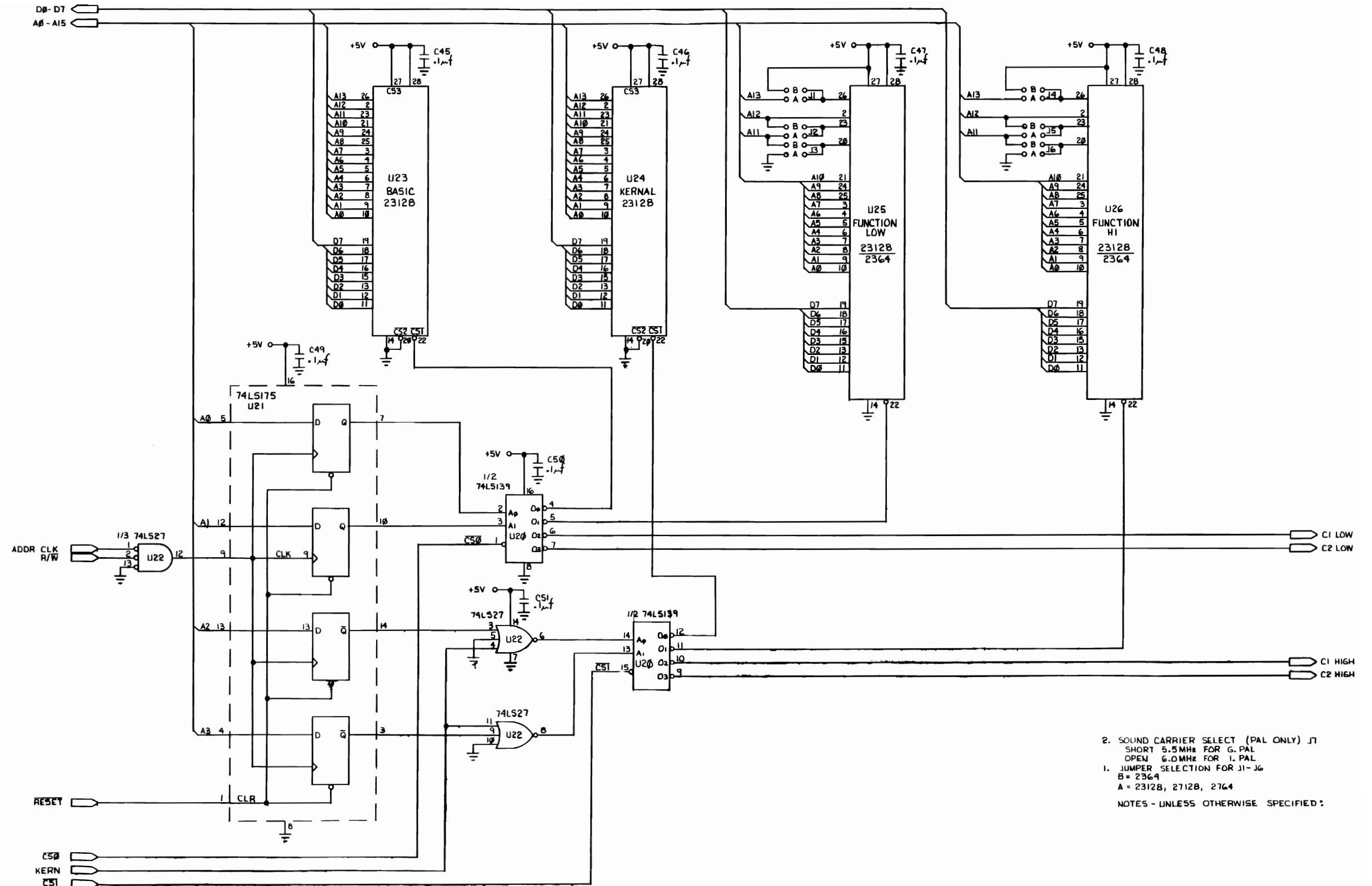
NC	1	28	VCC
A12	2	27	CS ₃
A7	3	26	A13
A6	4	25	A8
A5	5	24	A9
A4	6	23	A11
A3	7	22	CS ₁
A2	8	21	A10
A1	9	20	CS ₂
A0	10	19	D8
D1	11	18	D7
D2	12	17	D6
D3	13	16	D5
GND	14	15	D4

U23-318006-01 ROM – BASIC

PIN CONFIGURATION

NC	1	28	VCC
A12	2	27	CS ₃
A7	3	26	A13
A6	4	25	A8
A5	5	24	A9
A4	6	23	A11
A3	7	22	CS ₁
A2	8	21	A10
A1	9	20	CS ₂
A0	10	19	D8
D1	11	18	D7
D2	12	17	D6
D3	13	16	D5
GND	14	15	D4

U24-318005-04 ROM – KERNAL



Plus 4 Schematic #310164 (4 of 4)